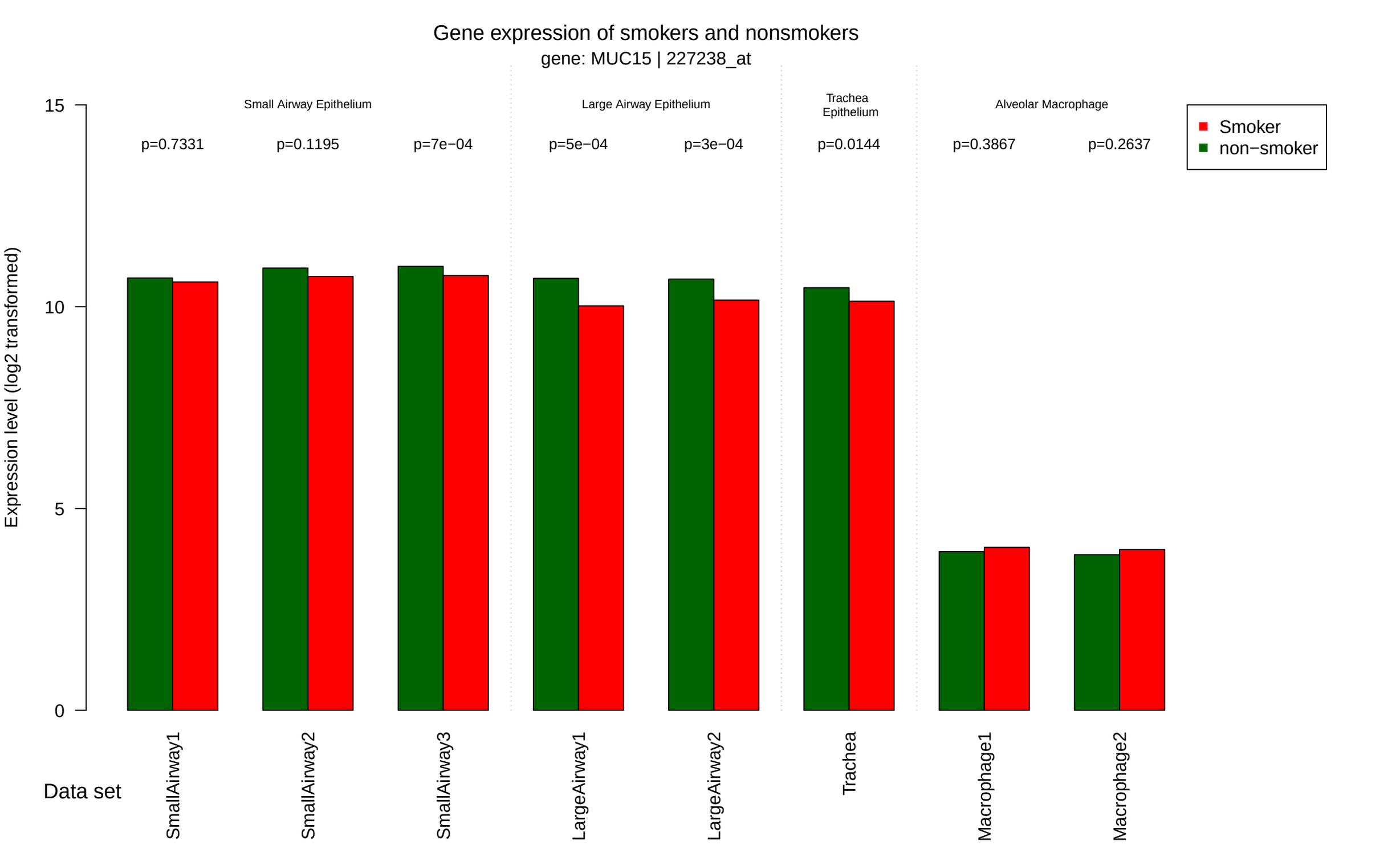
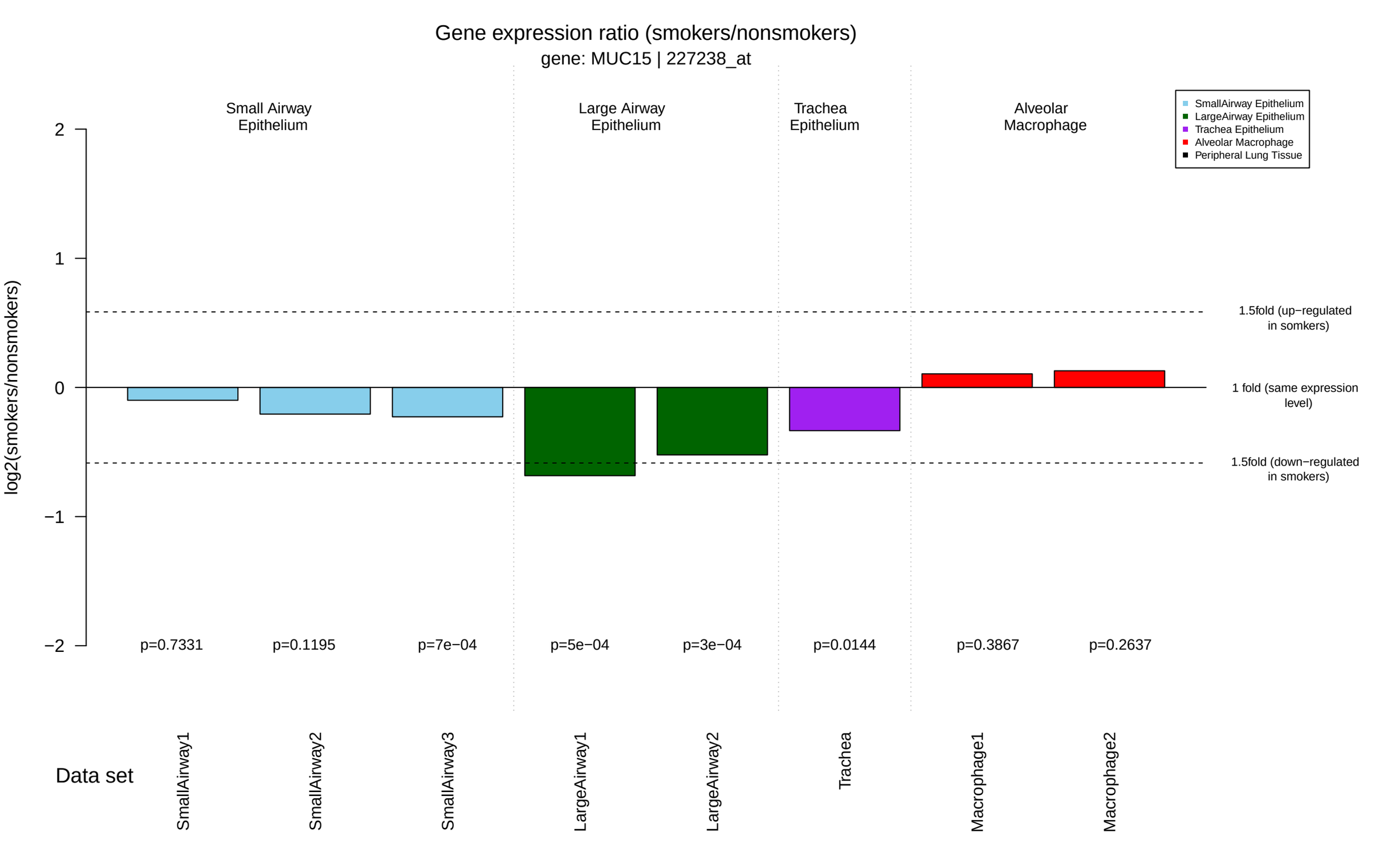
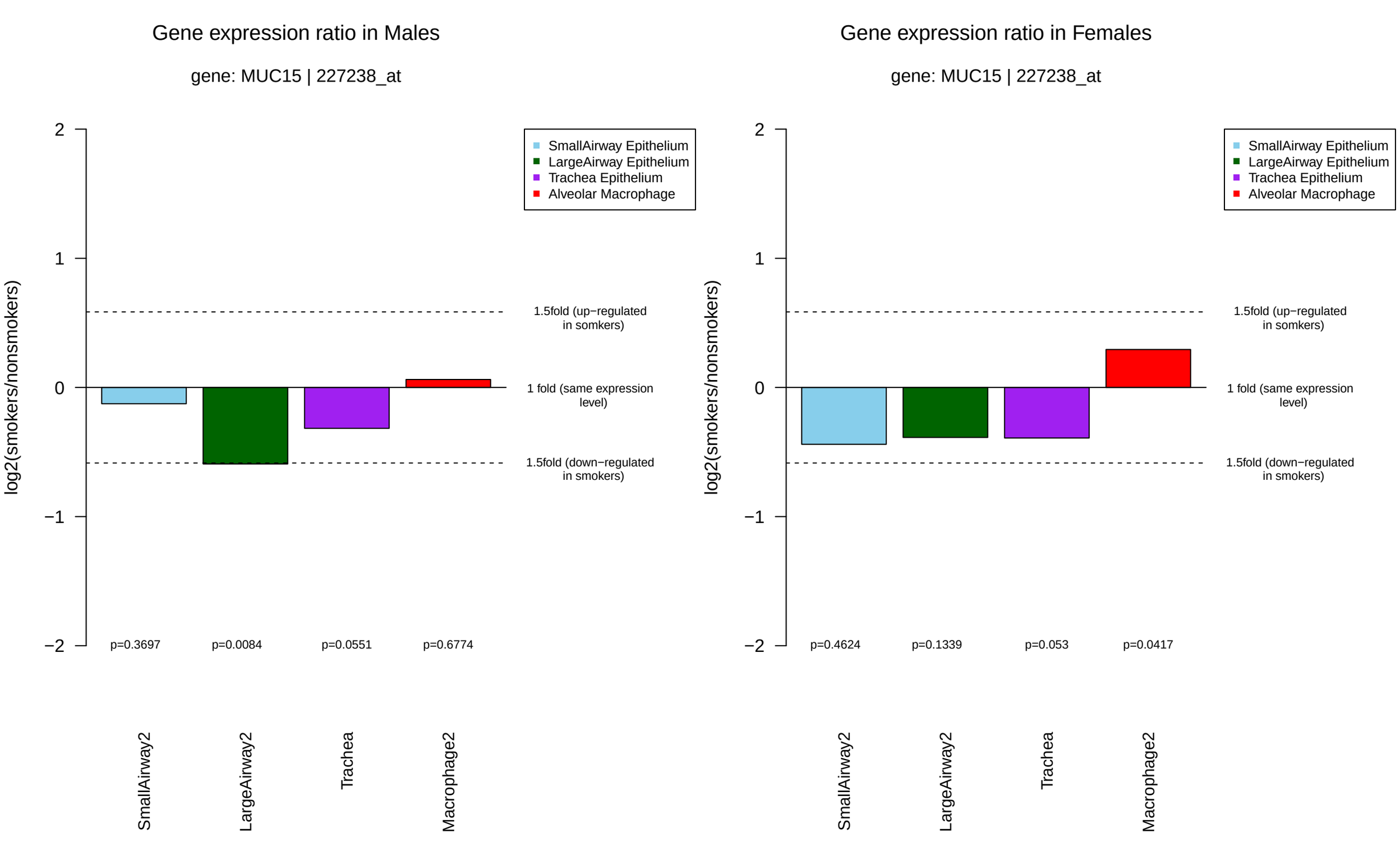
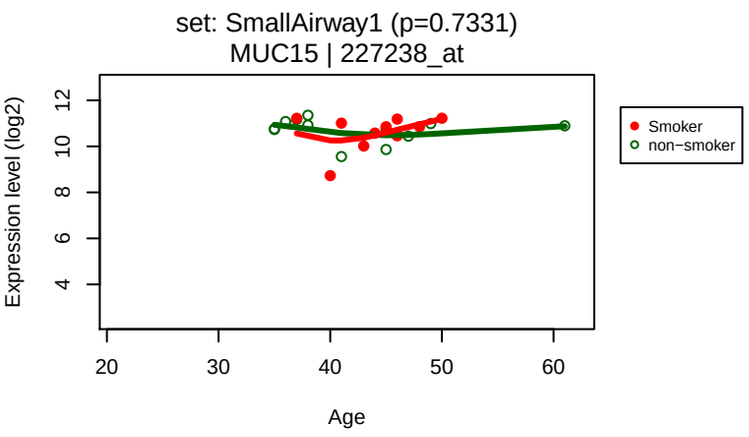
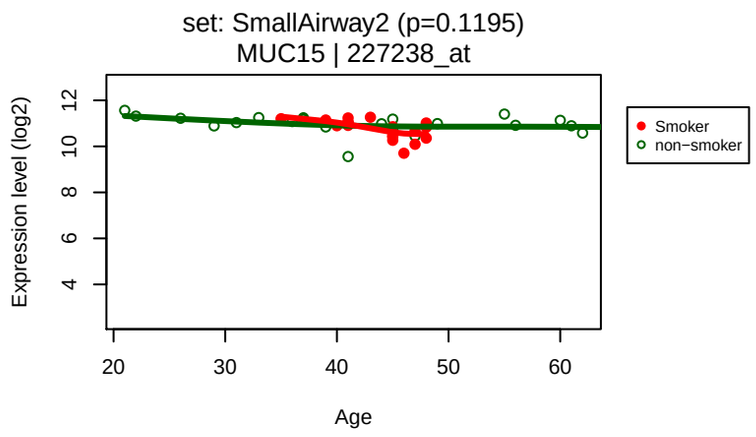
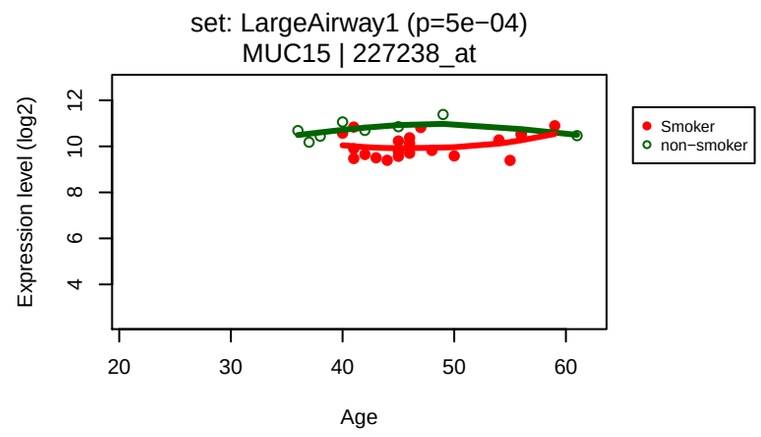

Gene expression of smokers and nonsmokers
gene: MUC15 | 227238_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7331
p=0.1195
p=7e−04
p=5e−04
p=3e−04
p=0.0144
p=0.3867
p=0.2637
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

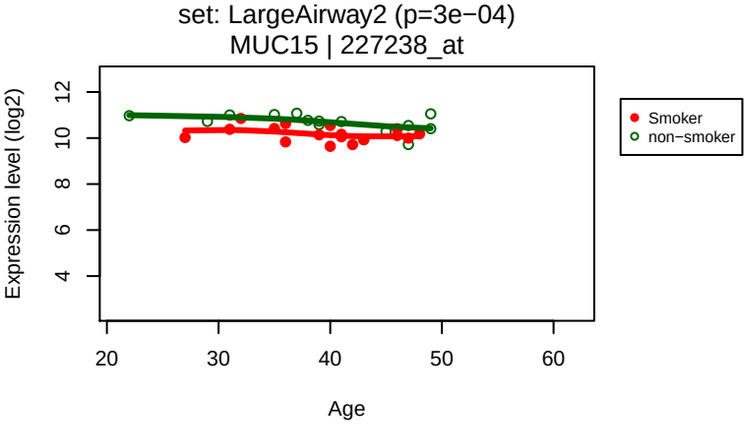

Gene expression ratio (smokers/nonsmokers)
gene: MUC15 | 227238_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7331
p=0.1195
p=7e−04
p=5e−04
p=3e−04
p=0.0144
p=0.3867
p=0.2637
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

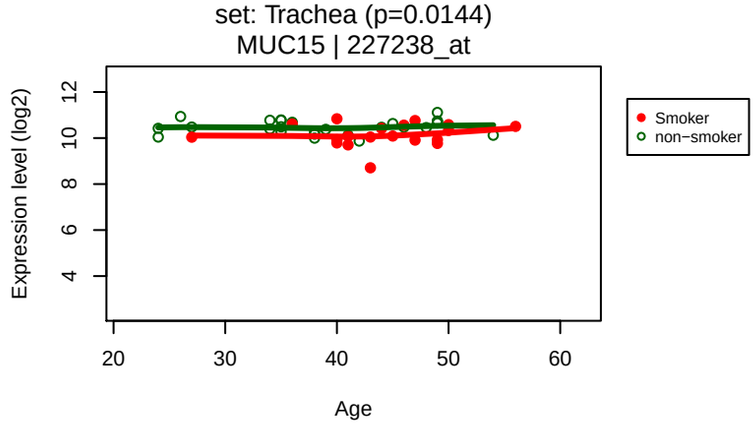

Gene expression ratio in Males
gene: MUC15 | 227238_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3697
p=0.0084
p=0.0551
p=0.6774
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MUC15 | 227238_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4624
p=0.1339
p=0.053
p=0.0417
SmallAirway2
LargeAirway2
Trachea
Macrophage2

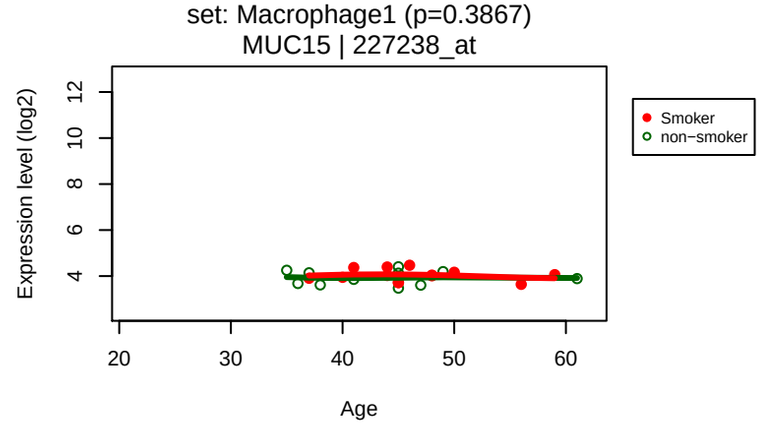

set: SmallAirway1 (p=0.7331)
MUC15 | 227238_at
Expression level (log2)
Age
Smoker
non−smoker

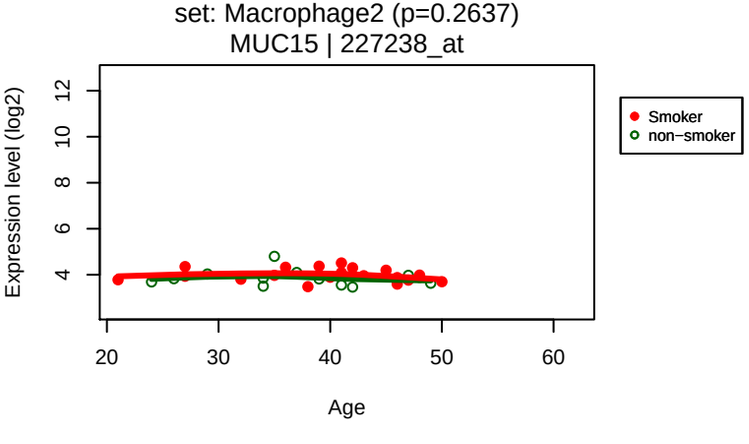

set: SmallAirway2 (p=0.1195)
MUC15 | 227238_at
Expression level (log2)
Age
Smoker
non−smoker

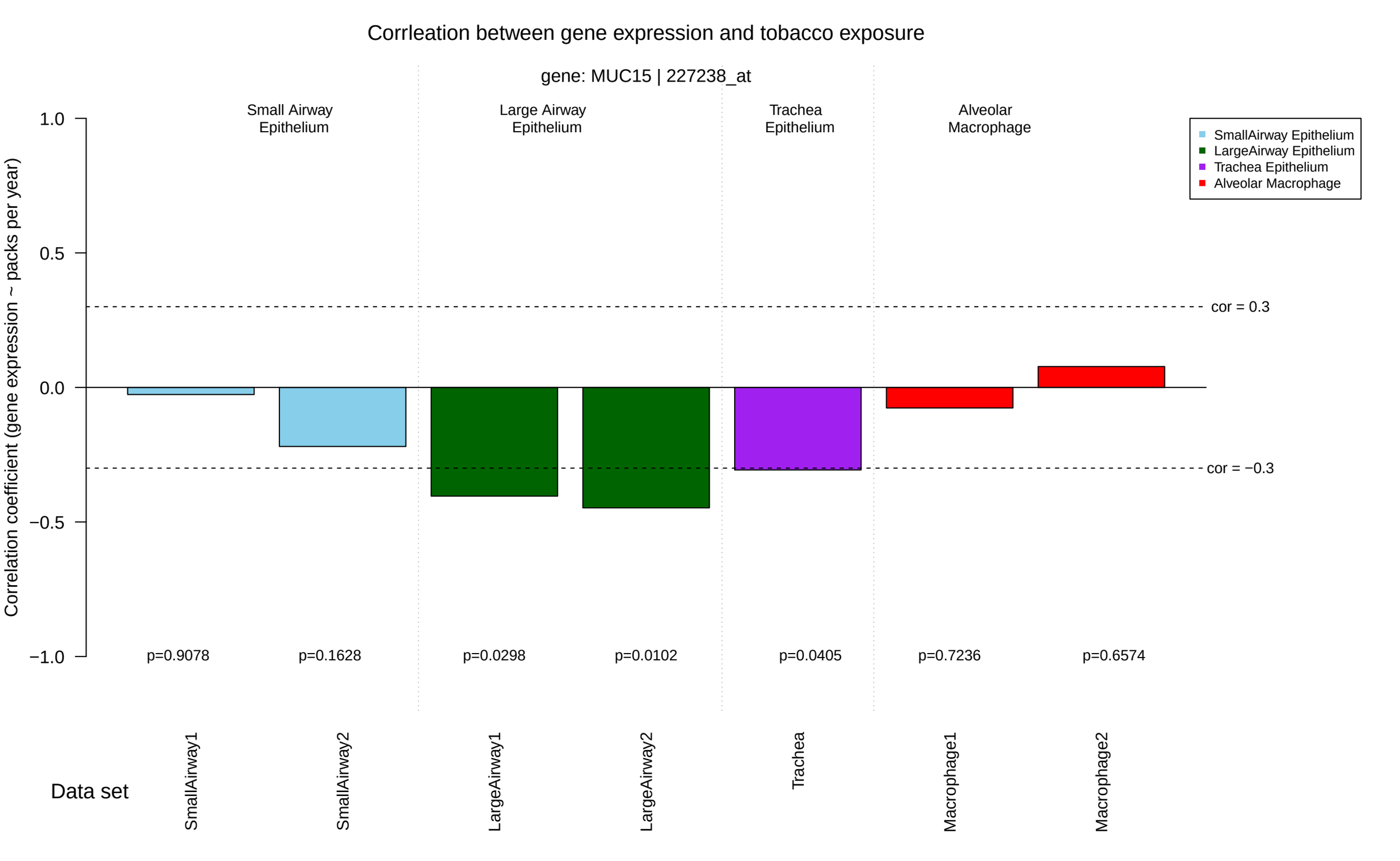

set: LargeAirway1 (p=5e−04)
MUC15 | 227238_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=3e−04)
MUC15 | 227238_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0144)
MUC15 | 227238_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3867)
MUC15 | 227238_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2637)
MUC15 | 227238_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MUC15 | 227238_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9078
p=0.1628
p=0.0298
p=0.0102
p=0.0405
p=0.7236
p=0.6574
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2